

1. Root Configuration & Deployment Files

pyproject.toml	Build & Test Config	Specifies package metadata, dependencies, and pytest asyncio disc
requirements.txt	Production Manifest	Pins production dependencies: FastAPI, PyMuPDF, Shapely, OpenCV,
.env / .env.example	Environment Config	Configures database URLs, SAFE_MODE=True, log levels, host/port,
Dockerfile	Container Image	Multi-stage Debian-slim container bundling Python 3.11, OpenCV ru
docker-compose.yml	Service Stack	Orchestrates SafeDig API server alongside local Ollama LLM contai
nginx_safedig.conf	Reverse Proxy	Nginx config handling SSL termination, 100MB body size for large
safedig.service	Systemd Daemon	Linux background service unit ensuring automatic restart and proc
safedig.db	Forensic Database	Async SQLite database storing immutable audit trails, SHA-256 has

2. Configuration & Domain Models (src/config/, src/domain/)

src/config/settings.py	Settings Singleton	Pydantic BaseSettings singleton validating environment confi
src/config/logging.py	Logging Subsystem	Structured enterprise logging with rotating file handlers an
src/domain/enums.py	Core Enums	Defines Decision, PDFModality, Severity, ReconciliationOutco
src/domain/index_record.py	Index Model	Pydantic model representing a statutory enquiry line from in
src/domain/document.py	Document Model	Represents physical utility map PDFs, tracking hashes, pages
src/domain/aoi.py	AOI Polygon	Encapsulates excavation Area of Interest polygon, vertices,
src/domain/legend.py	Legend Profile	Defines cartographic color bands, line stroke styles, and sy
src/domain/reconciliation.py	Reconciliation Result	Holds 2D geometric intersection calculations between AOI and
src/domain/policy.py	Policy Gate Models	Data models for individual GateCheck items and overall Polic

3. Ingestion, Spatial Math & Computer Vision (src/spatial/, src/vision/)

src/ingestion/inventory.py	Inventory Matcher	Fuzzy token matching linking index records to authorita
src/index/parser.py	Index Parser	Parses statutory index.xlsx sheets with self-healing cr
src/spatial/aoi_detector.py	AOI Extraction	Scans drawings for red boundary lines, hatching, and co
src/spatial/vector_parser.py	Vector Engine	PyMuPDF path extractor parsing line strokes, widths, an
src/spatial/reconciler.py	Spatial Reconciler	Computes exact 2D polygon intersections between AOI buf
src/spatial/geometry_utils.py	Geometry Math	Shapely helper functions for coordinate transforms and
src/vision/detector.py	Vision Coordinator	Dual-engine coordinator fusing vector graphics and rast
src/vision/rasterizer.py	Map Rasterizer	Renders high-resolution 300 DPI OpenCV BGR image arrays
src/vision/color_filter.py	Color Filter	HSV/RGB color masking isolating utility lines per provi
src/vision/contour_analyzer.py	Contour Engine	Morphological analysis identifying point symbols (valve

4. Safety Policy Engine & Advisory LLM (src/policy/, src/llm/)

src/policy/gates.py	17 Release Gates	Evaluates all 17 mandatory release gates ensuring zero
src/policy/engine.py	Policy Engine	Hierarchical decision ladder routing to AUTO_CLEAR, HUM
src/llm/client.py	Ollama Client	Asynchronous HTTP client interfacing with local Ollama
src/llm/prompts.py	System Prompts	Few-shot safety-critical prompts directing LLM discrepa
src/llm/explainer.py	Advisory Explainer	Synthesizes operator notes with 0ms deterministic rule

5. Orchestration, QA Workspace & REST API (src/orchestration/, src/api/)

src/orchestration/graph.py	LangGraph DAG	Assembles stateful pipeline graph connecting ingestion,
src/orchestration/nodes.py	Pipeline Nodes	Execution logic for individual graph nodes with failure
src/qa/workspace.py	QA Workspace Service	Assembles workspace models, renders visual crops, handl
src/api/main.py	FastAPI App Entry	FastAPI entry point configuring CORS, static routes, an
src/api/routes/jobs.py	Jobs API	Endpoints for job ingestion, listing, status checking,
src/api/routes/qa.py	QA Workspace API	Serves workspace state, document metadata, and receives
src/api/routes/evidence.py	Evidence API	Streams high-resolution AOI crops, whole sheet plans, a
src/api/static/index.html	Web Console HTML	Single-page dashboard with animated radar scanner and r
src/api/static/app.js	Console Frontend JS	Client app logic: canvas pan/zoom, gate rendering, and
src/utils/paths.py	Path Resolution	Self-healing path resolver fixing cross-machine path di
src/pipeline.py	Pipeline CLI Entry	Standalone execution script for processing jobs via ter